Project 12 — Explained Simply (For a 3rd Sem Student)

🎯 The One-Line Gist

> **"An AI that not only makes decisions, but tells you *why* it made them — and *what you can do* to get a better result — using actual cause-and-effect logic, not just pattern-matching."**

🧠 Start Here: The Real-World Story

Imagine you apply for a **bank loan**. The bank's AI system rejects you.

You ask: **"Why was I rejected?"**

The bank says: *"Our AI said no."*

You ask: **"Okay, what can I change to get approved?"**

The bank says: *"We don't know."*

😤 That's the problem this project solves.

🔍 Breaking It Down — Three Core Problems

Problem 1: Current AI is a Black Box

Most AI models (like the ones used by banks, hospitals, companies) just give you an output — **approved / rejected, high risk / low risk** — with no explanation.

Problem 2: Existing "Explanation" Tools Are Misleading

There are tools like **SHAP** and **LIME** that try to explain AI decisions. But here's the catch:

> They tell you what *correlated* with the decision — not what *caused* it.

Simple analogy:

> Ice cream sales and drowning incidents both go up in summer. They're *correlated*. But eating ice cream doesn't *cause* drowning. Summer heat causes both.

SHAP would say "ice cream sales is an important feature." Your project says — *no, let's find the actual cause.*

Problem 3: "What Should I Do?" Has No Answer

Even if an AI says "you were rejected because of low income," it doesn't tell you:

- *By how much do I need to increase my income?*
- *What's the minimum realistic change I need to make?*
- *Is there something else I can change instead?*

This is called **Actionable Recourse** — giving the user a realistic path to a better outcome.

💡 What Your Project Does (The Solution)

Your system does **3 things** that existing tools don't:

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1. CAUSAL ATTRIBUTION  →  Finds what actually CAUSED the decision
2. ACTIONABLE RECOURSE →  Tells you what you can REALISTICALLY CHANGE
3. DECISION DASHBOARD  →  Shows all of this in a simple, visual interface
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🧩 Key Concepts You Need to Understand

1. Correlation vs Causation ⭐ (Most Important)

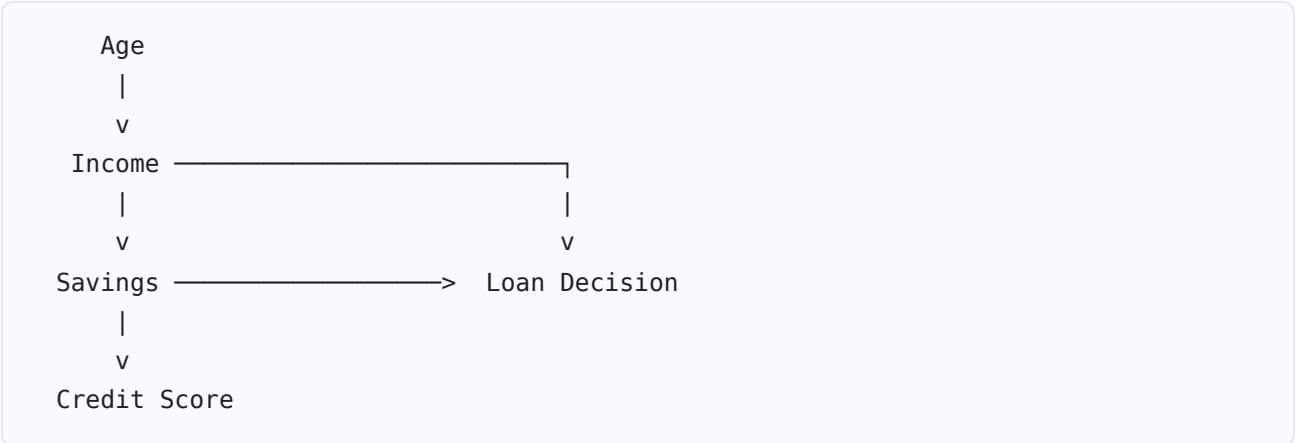
	Correlation	Causation
What it means	Two things happen together	One thing directly causes another
Example	Shoe size & reading ability in kids (both increase with age)	Studying → Better marks
Tool used	SHAP, LIME	Your project (SCM + do-calculus)
Problem	Can be misleading	Accurate & trustworthy

How to explain in viva:

> *"Sir, SHAP finds features that are statistically associated with the output. But correlation ≠ causation. Our system uses Structural Causal Models to find features that, when you actually change them, will genuinely change the outcome."*

2. Structural Causal Model (SCM) — The "Map of Cause & Effect"

Think of it as a **family tree, but for causes.**



This diagram (called a **DAG — Directed Acyclic Graph**) shows:

- Income affects Savings
- Savings and Income together affect the Loan Decision

- Age affects Income (older = more experience = higher income possibly)

Once you have this map, you can ask: *"If I change Income, what actually happens to the Loan Decision — following the causal chain?"*

How to explain in viva:

> *"Sir, we build a DAG — a Directed Acyclic Graph — that captures cause-effect relationships between features. This is called a Structural Causal Model. It lets us compute the true effect of changing one variable on the final decision."*

3. Do-Calculus — The Math Behind "What If?"

Created by **Judea Pearl** (Turing Award winner — basically the godfather of causality in AI).

Normal statistics asks: *"What is P(Loan=Yes | Income=High)?"* — probability given observation.

Do-calculus asks: *"What is P(Loan=Yes | **do**(Income=High))?"* — probability if we **force/intervene** to make income high.

The difference: In `do()`, you're not just observing high-income people. You're asking — *if someone's income **were** made high by an intervention, what would happen?*

Simple analogy:

> Observing: *People who go to hospitals tend to die more* → correlation. Do-calculus: *If we force everyone to go to hospital, what happens?* → actual causal effect (they live longer).

4. Actionable Recourse — "What Can I Actually Change?"

Your system generates suggestions like:

Feature	Current Value	Suggested Change	Cost
Monthly Savings	₹2,000	₹5,000 (+₹3,000)	Medium
Credit Score	580	650 (+70 points)	Low effort over 6 months
Age	22	❌ Cannot change	Immutable

Key rules your system follows:

- **Mutable features only** — can't tell someone to change their age/gender
- **Causally consistent** — if you change savings, the system also updates its estimate of credit score (because they're causally linked)
- **Minimum cost** — finds the *smallest* change that flips the decision

5. The System Architecture (4 Modules)

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RAW DATA (loan application, patient record, etc.)
     ↓
MODULE 1: Clean the data, label which features can be changed
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MODULE 2: Learn the cause-effect map (DAG) + compute causal attributions
     ↓
MODULE 3: Generate "what you can change" suggestions (recourse)
     ↓
MODULE 4: Show everything in a nice dashboard to the analyst/user
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🗣️ How to Explain It in Presentation / Viva

Opening Line:

> *"Sir/Ma'am, most AI systems today make high-stakes decisions — in banking, healthcare, hiring — but cannot explain those decisions in a trustworthy way. Tools like SHAP and LIME exist, but they show correlations, not causes. Our project builds a causal explainability system that identifies the true drivers of any AI decision and provides the user with realistic, actionable steps to achieve a better outcome."*

If Asked "What is the novelty?":

> *"The novelty is two-fold. First, we use Structural Causal Models instead of correlational attribution, so explanations are causally valid. Second, we generate actionable recourse — specific, realistic steps the user can take — filtered to only include features they can actually change, while respecting the causal relationships between those features."*

If Asked "What is SHAP's limitation?":

> *"SHAP is based on Shapley values from cooperative game theory. It measures the average marginal contribution of each feature to the prediction. But this is a correlational measure — it tells you which features were associated with the prediction, not which features caused it. In the presence of confounders or causal dependencies, SHAP can highlight the wrong features entirely."*

If Asked "What is a DAG?":

> *"A Directed Acyclic Graph is a graph where nodes represent variables and directed edges represent causal relationships. Acyclic means there are no loops — a variable cannot cause itself indirectly. We use this structure as the backbone of our Structural Causal Model."*

If Asked "What is do-calculus?":

> *"Do-calculus, developed by Judea Pearl, is a mathematical framework for computing the effect of interventions — not just observations. The notation do(X=x) means we forcibly set variable X to value x and observe what happens to the outcome, simulating a real-world intervention."*

📚 Things You Should Know Cold (For Viva)

Topic	What to Know
Correlation vs Causation	Definition + 1 clear example
DAG	What it is, how to draw a simple one
SCM	It's a DAG + equations describing how each variable is generated
SHAP	What it does, why it's correlational, its limitation
LIME	Local surrogate model, also correlational
do-calculus	Conceptual meaning — "effect of intervention"
Actionable Recourse	Realistic suggestions, mutable features only
XGBoost/LightGBM	The underlying ML model doing the prediction
FastAPI	Python framework to build backend APIs
Streamlit	Python library to build quick dashboards

🔑 The 3-Sentence Summary (Memorize This)

> *"Our project addresses the problem of unexplainable and non-actionable AI decisions in high-stakes domains like banking and healthcare. Unlike existing tools such as SHAP and LIME — which show correlations — our system builds a Structural Causal Model to identify the true causal drivers of each decision. It then generates causally feasible, actionable recourse — specific and realistic steps the user can take to change an adverse outcome — presented through an interactive decision-support dashboard."*

> TIP
>
> In your viva, always connect your technical answer back to the real-world problem (loan rejection, medical diagnosis, hiring). Examiners love when you can bridge theory and application.